

Delay Time cont'd. Voices

These parameters set the delay time, in seconds, for each individual voice. The range for each algorithm is shown below.

You can set and display delay values in units of time, or with tempo values. Press **Up** and **Tempo** simultaneously to toggle between these two options. When time units are selected, delay times are set and displayed in milliseconds (From 0ms to the maximum available delay for that parameter). When tempo values are selected, delay values are set and displayed as a ratio of echoes to beats (from 24:1 to 1:24). This will automatically synchronize the delay to the current tempo (MIDI, Internal, or TAP — see *Tempo Mode* in Chapter 2.) For example, a delay setting of 1:2 (1 echo for every 2 beats) will produce half-note delay rhythms synchronized to the current tempo.

Algorithm	Delay Time
Concert Hall	0-1.365 sec
Plate	0-1.365 sec
Chamber	0-1.365 sec
Inverse	0-1.365 sec
Infinite	0-1.365 sec
Glide>Hall	0-10.581 sec
Chorus+Rvb	0-1.365 sec
M-Band+Rvb	0-10.922 sec
Res 1>Plate	0-9.557 sec
Res 2>Plate	0-9.557 sec

When display of values in BPM has been selected, these are set as fractions of a beat. The smallest fraction is 1/24th beat. Changing these values live can produce a variety of pitch shift, and flange effects. Delay times can also be tempo modulated.

Parameters in this row provide level and phase setting for feedback of individual voices, as well as a master feedback parameter for all voices. In the Glide>Hall algorithm, a duplicate set of parameters is included for cross-feedback.

Feedback/ Cross Feedback

Master

A Master Feedback control is available in effects which have a feedback level control for each voice. It allows the feedback level for all voices to be simultaneously adjusted over a range of 0-100%.

Voices

In the 4-Voice algorithms, four voice controls are provided for adjustment of the level and polarity of signals recirculated back to the input of individual voices. Note that, as Voices 1 and 2 and Voices 3 and 4 are paired, caution should be observed not to create feedback within a pair which exceeds more than 100%.

In the Chorus+Rvb and the M-Band+Rvb algorithms, six voice parameters control the feedback level of the voice delays. Voices 1, 2, and 3 Fbk control the individual voice feedback levels from the left channel voice delay outputs to the left channel delay feedback input. Voices 4, 5, and 6 Fbk control the individual voice feedback levels from the right channel voice delay outputs to the right channel delay feedback input.

The range of these parameters is from +100% down to 1% (Phase Normal feedback), 0% (Off), -1% up to -100% (Phase Inverted feedback).

In the Glide>Hall algorithm, the Feedback row provides both Feedback and Cross Feedback parameters as described below. Press **Load/★** at each row position to toggle between Feedback and Cross Feedback parameters.

Mstr Fbk, Mstr X-Fbk

These parameters allow the feedback and cross feedback levels for all voices to be simultaneously scaled over a range of 0-100%.

Voices 1-6 Fbk, X-Fbk

These parameters control the feedback and cross feedback levels and the polarity of the voice delays.

Voices 1, 2, and 3 Fbk control the individual voice feedback levels from the left channel voice delay outputs to the left channel delay feedback input.

Voices 1, 2, and 3 X-Fbk control the individual voice feedback levels from the left channel voice delay outputs to the right channel delay feedback input.

Voices 4, 5, and 6 Fbk control the individual voice feedback levels from the right channel voice delay outputs to the right channel delay feedback input.

Voices 4, 5, and 6 X-Fbk control the individual voice feedback levels from the right channel voice delay outputs to the left channel delay feedback input. The range of these parameters is from +100% down to 1% (Phase Normal feedback), 0% (off), -1% up to -100% (Phase Inverted feedback).

The sum of all Fbk and X-Fbk values for each channel should be less than 100%.

Filters In the M-Band+Rvb algorithm, Row 5 contains parameters for cut-off frequencies of low and high cut filters for each voice, as well as master low and high cut controls for all voices.

Mstr LC/HC

Two master controls are provided in the first row position. Press **Load/★** to toggle between them. The High Cut Master allows simultaneous adjustment of the high frequency cut-off points for the filters on all voices in single semitone increments up or down from the individual voice settings. The Low Cut Master allows simultaneous adjustment of the low frequency cut off points for the filters on all voices in single semitone increments up or down from the individual voice settings.

V1-V6 LoCut and HiCut

The low cut and high cut filters operate at 12dB/octave. There are individual low cut and high cut filters for each of six voices. The ranges of the filter cut-off points are from 20-20,000Hz at approximately semitone intervals across the entire range for a total of 121 cut off points. The low cut filters have an OFF position just below 20Hz. The high cut filters have an OFF position just above 20,000Hz.

Glide FX In the Glide>Hall algorithm, Row 3 contains level, delay and feedback parameters for a pair of stereo gliding delays that precede the 6-voice delay.

Gld Lvl

Gld Lvl controls the stereo output level from all gliding delays (A, B, Left and Right). It has a range of Full (0dB) down to -85dB and Off.

A Left (A Lvl L and A Dly L)

These parameters control the output level and delay time of the left channel A Glide Delay. A Lvl L has a range of Full (+0dB), Phase Normal down to -85dB, Phase Normal, OFF, -85dB Phase Inverted up to Full (+0dB), Phase Inverted. A Dly L has a range of 0.0 milliseconds to 42.0 milliseconds in 0.1 millisecond units.

A Right (A Lvl R and A Dly R)

These parameters control the output level and delay time of the right channel A Glide Delay. They have the same ranges as A Left.

B Left (B Lvl L and B Dly L)

These parameters control the output level and delay time of the left channel B Glide Delay. They have the same ranges as A Left.

B Right (B Lvl R and B Dly R)

These parameters control the output level and delay time of the right channel B Glide Delay. They have the same ranges as A Left.

Fbk L/R

These parameters control the feedback levels of the corresponding left and right channel Glide Delays. The range of these parameters is from +100% down to 1% (Phase Normal feedback), 0% (off), -1% up to -100% (Phase Inverted feedback).

X-Fbk L/R

These parameters control the corresponding cross feedback levels of the left and right channel glide delays. Specifically, X-Fbk L controls the feedback from the left channel A+B glide delay output to the right channel glide delay feedback input. X-Fbk R controls the feedback from the right channel A+B glide delay output to the left channel glide delay feedback input. The range of these parameters is from +100% down to 1% (Phase Normal feedback), 0% (off), -1% up to -100% (Phase Inverted feedback).

Glide FX cont'd.

In each algorithm, the Levels row contains parameters for the phase and level of each voice, as well as a master level parameter for all voices.

Levels**Master**

The Master Level controls the overall level in effects which have more than one voice. This permits simultaneous changes in all voice levels without altering the balance between individual voices.

In the Res1>Plate and Res2>Plate algorithms, the Master parameter contains two master parameters for the six resonator voices: Mstr Lvl and Mstr Fbk. (**Load/★** will toggle between them.) Mstr Lvl is a master level control for all of the resonator voices, as described above. Use care when adjusting this value. The resonators will put out high levels if the input source has any components that are close in pitch to the resonator tunings. In general, this control should be set to 0dB or less.

The Mstr Fbk control allows the feedback levels for all voices to be scaled over a range of 0-100%.

Voices

Individual controls are provided for each Voice Level. Voices can be turned completely off, full on, or set to any point between, in 1dB increments. There are as many voice level controls as there are voices in an effect.

In the Res1>Plate and Res2>Plate algorithms, positions 1-6 contain both level and feedback parameters for each of the six resonator voices. (**Load/★** will toggle between them.) Lvl controls the individual level and polarity for each voice. Voices can be turned completely off, full on, or set to any point in between in 1dB increments. Use care when adjusting this value. The resonators will put out high levels if the input source has any components that are close in pitch to the resonator tunings. In general, this control should be set to a value less than Full.

The Fbk controls set the feedback level and polarity of each voice. Voices 1-3 Fbk control the feedback levels from the left channel delay voice outputs to the left channel delay voice inputs. Voices 4-6 Fbk control the feedback levels from the right channel delay voice outputs to the right channel delay voice inputs. The sum of the values for either side should be less than 100%. (Disregard the negative sign when adding values.)

Modulation The Modulation row, which is the same for every algorithm, contains the parameters for the PCM 81's internal modulation sources. Use the Patch row to assign these modulators to any PCM 81 effect parameter.

Mod: LFO

Four parameters are available: **Shape**, **P Width**, **Depth**, and **Rate**.

Shape allows you to select the wave shape which will be used when "LFO" is selected as a patch Source. The choices are:

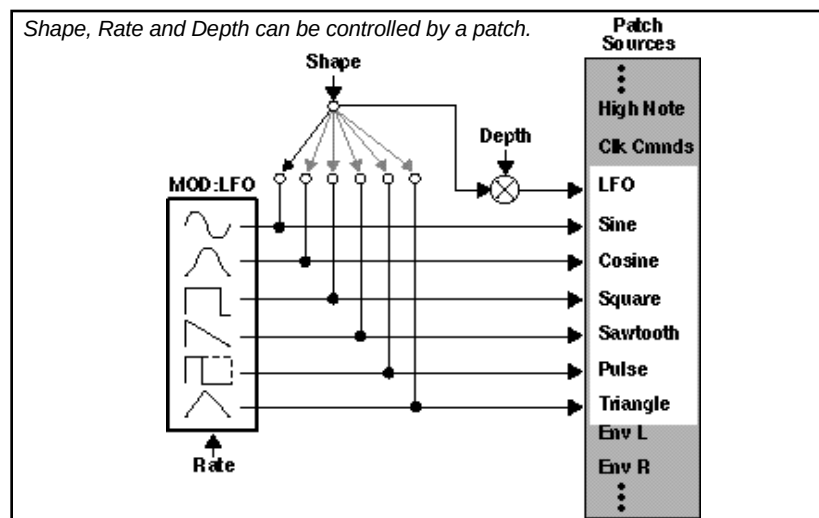
Sine
Cosine
Square
Sawtooth
Pulse
Triangle

P Width determines the proportion of each pulse wave cycle for which the LFO is on (1-99%). For example, setting **P Width** to 50% means that the LFO is on for half of its cycle. The effect of this control will only be heard if you are using the Pulse shape.

Depth scales the output of the LFO from 0 to 100%.

Rate sets the speed (0-25Hz) at which the LFO cycles. It can be set in time values (such as 1.5Hz) or in tempo values (such as 3:2 cycles per beat). Simultaneously pressing **Up** and **Tempo** will toggle these two display options.

Note: The PCM 81 allows six LFO shapes (sine, cosine, sawtooth, triangle, square and pulse) to be selected as patch Sources, as well as the LFO itself. All of these are generated by a single LFO, and are controlled by the single **Rate** control. When "LFO" is selected as a patch Source, the output will be the **Shape** selected here. The amplitude of the LFO output is controlled by **Depth**. Both Shape and Depth are available as patch Destinations and can be controlled externally. Shape, Pulse Width, Depth, and Rate are all available as patch Destination parameters, and can be controlled externally. See *Patching* in Chapter 2.



Mod: AR Env

This envelope generator's output, when turned on, will go from 0 to 127. How quickly it goes from 0 to 127 is determined by the setting of **Attack** (0-10 seconds). Once the envelope generator has reached 127, it remains there as long as it is turned on. When it is turned off, it goes from 127 to 0, at the rate determined by **Release** (0 to 10 seconds).

T Src allows you to select a Source to turn the envelope generator on and off.

T Lvl allows you to select a specific threshold value which the **T Src** must reach to turn the envelope generator on.

The **Mode** parameter allows you to determine the behavior of the envelope controller in relation to the threshold value. Four settings are available: **Repeat**, **One Shot**, **Retrigger**, and **Off**.

- Repeat** As long as the threshold source remains at or above **T Lvl**, the envelope cycles from attack to release. If A=R, the output of the envelope is a triangle wave.
- One Shot** Once **T Lvl** is reached, the envelope will go through its entire attack cycle. Once the attack cycle is completed, if Source value is *below* **T Lvl**, the envelope will immediately fall at the specified **Release** rate. The envelope will go through its entire release cycle, even if the source subsequently rises above **T Lvl**. If the Source value is *at or above* **T Lvl**, the envelope will remain at 127 until the **T Src** falls.
- Retrigger** As long as the level is at or above **T Lvl**, the envelope will go through its attack cycle. If the level falls below **T Lvl** before attack is completed, the envelope immediately begins to release. Likewise, if the **T Lvl** is crossed again before the release is completed, the attack cycle will begin again.
- Off** This control turns the AR envelope off (and frees up processor time). To optimize PCM 81 real-time response, set AR Env to **Off** when it is not being used.

Mod: Env L and Mod: Env R

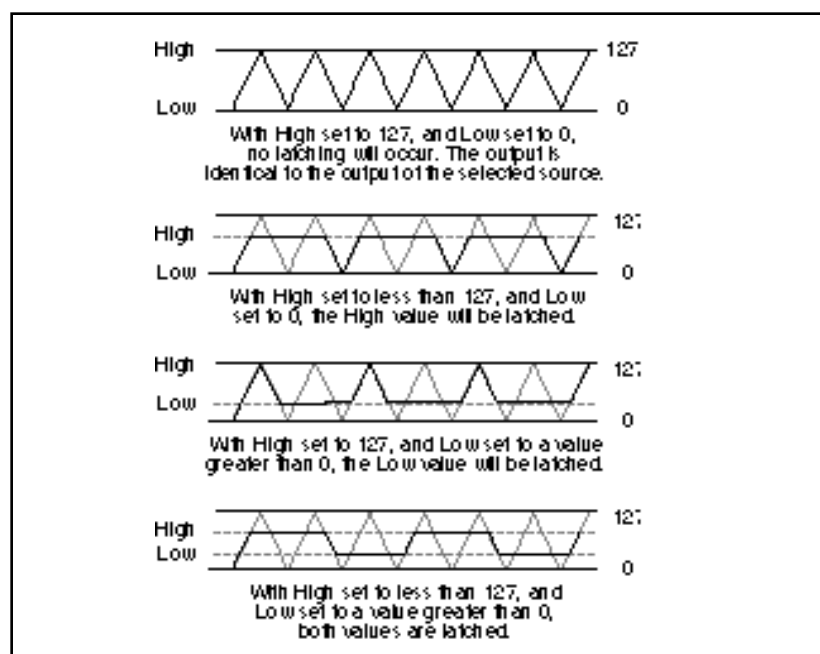
These are left and right input signal envelope followers. The only available parameter is **Release** which is set in milliseconds. This control allows you to specify the release rate (0-10 seconds) when the input level drops.

Modulation cont'd.

Modulation cont'd. Mod: Latch

The latch is a very flexible modulation source. It can be used to do such things as derive a switch from a continuous "return to zero" source (like MIDI After Touch). It can turn a momentary (on/off) footswitch into a latching footswitch (push on/ push off), and it can divide the switching rates of sources in half or thirds.

The latch has three parameters: **Src**, **High** and **Low**. Any patch source can be the **Src** (See Source listing under Patching.) **High** and **Low** are threshold values. The latch works as follows:



There is no output from the latch until the **Src** value falls within the range defined by the settings of **High** and **Low**. While the source value is between these thresholds, the output of the latch is the same value as the source. When the source value reaches or passes either threshold, the output of the latch holds at the limit value until the source value passes through the threshold twice. The latch can be set to hold at either the low threshold, the high threshold, or both. Setting **Low** to 0 disables latching at the low threshold. Setting **High** to 127 disables latching at the high threshold.

See preset **P0 3.0 FSw2 Rotary** as an example.

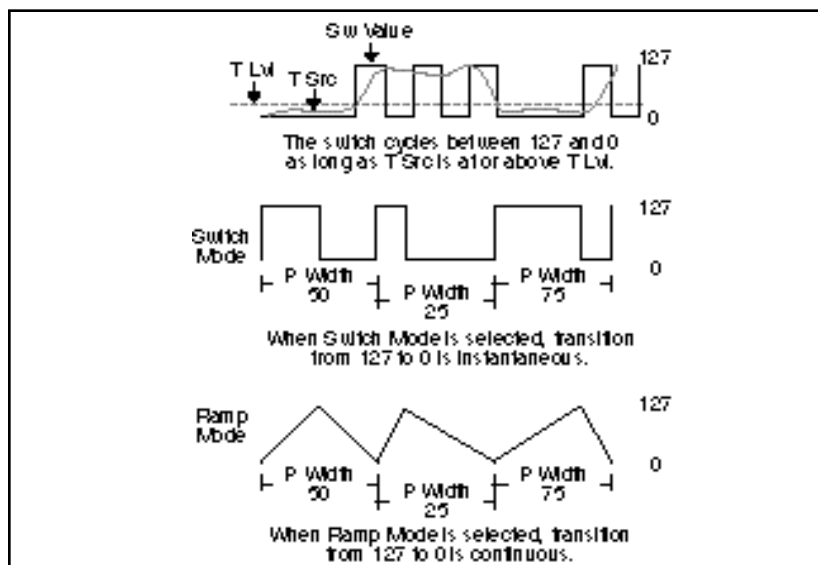
Mod: Sw 1 and Mod: Sw 2

These are identical time switches. Each has five parameters: **Rate**, **P Width**, **Mode**, **T Lvl** and **T Src**.

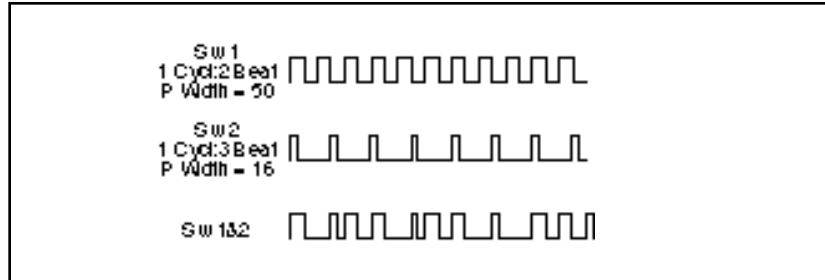
Modulation cont'd.

- Rate** sets the speed at which the switch cycles. It can be set in time values (such as 1.5 Hz) or tempo values (such as 3:2 Cycles per Beat). Simultaneously pressing **Up** and **Tempo** will toggle these display options.
- P Width** determines the proportion of each switch cycle during which the switch is on. For example, setting **P Width** to 50% means that the switch is on for one-half of a cycle.
- Mode** determines the “shape” of the switch output. Three settings are available: **Switch**, **Ramp**, and **Off**. When **Switch** is selected, the transition from on to off is instantaneous, i.e. the switch output resembles a pulse wave. When **Ramp** is selected, the transition from on to off is continuous, i.e. the switch output resembles a triangle or sawtooth wave.
- Off** turns the switch off (and frees up processor time). To optimize PCM 81 real-time response, set switch to **Off** when it is not being used.
- T Lvl** sets the threshold value at which the switch will begin to cycle.
- T Src** selects a patch source to drive the switch. The output of the switch is 0 as long as **T Src** is set to a value below **T Lvl**. Once the source value reaches or passes **T Lvl**, the switch will begin to cycle between on (127) and off (0) at the speed set by **Rate**.

Note that both Rate and P Width are available as patch Destinations, allowing them to be dynamically controlled by other patch sources. Switches are reset to the beginning of their cycles whenever **Tap** is pressed.



Modulation cont'd. A special, composite output of these switches, called **Sw 1&2** is available as a patch source. The value of **Sw 1&2** alternates between the output of Sw 1 and the output of Sw 2. The alternation occurs on the transition from on to off. Note that both Sw 1 and Sw 2 must be active for the alternation to occur.



Panning Parameters in the Panning row provide control of panning of individual effects voices, as well as a master panning parameter for all voices.

Master

The Pan Master provides simultaneous control over the panning of all voices in the effect. The range of the Pan Master is 50L (full left) through 1L, C (Center), 1R through 50R (full right) for a total of 101 positions. The Master Pan is additive to the individual voice panners such that, if a voice is set to L10, and the Master is set to 20R, the voice will be at 10R.

Voices

Individual pan controls are provided to position the output of each voice. The range of each voice is 50L (full left) through 1L, C (Center), 1R through 50 R(full right) for a total of 101 positions.

The last row of each algorithm's edit matrix is the Patch row. This row provides parameters for creating as many as ten patches in each effect. Each row position (Patch 0-9) has three controls available: **Src**, **Dst**, and **Values**. Press **Load/✱** to cycle among these selections.

Patching

Src

Use ADJUST to select any of the sources listed below.

Dst

Use ADJUST to select any effect parameter except those on the Patch row.

Values

Use ADJUST to assign Destination values to specific Source values. These assignments are made in pairs, each with a value for the Source and a value for the Destination. For example, the default is two pairs mapped as follows:

- minimum Source value (0) = minimum Destination value
- maximum Source value (127) = maximum Destination value

This gives you a linear relationship between the parameter and the controller. Inverse control is accomplished easily by reversing these settings. As many as eight pairs of Destination/ Source values can be assigned here, providing an exciting new level of dynamic control.

See Chapter 2 for a complete description of the Patching System.

PCM 81 Patch Sources			
Internal		MIDI Controller Numbers	
LFO		(PCM 81 interprets 000 as Bank Select)	070 Sound Var
Sine		001 Mod Wheel	071 Timbre
Cosine		002 Breath	072 Release
Square		003 Ctl 3	073 Attack
Sawtooth		004 Foot Ctl	074 Bright
Pulse		005 PortaTime	075 Sound 6
Triangle		006 Data Entry	076 Sound 7
Env L		007 Volume	077 Sound 8
Env R		008 Balance	078 Sound 9
AR Env		009 Ctl 9	079 Sound 10
Latch		010 Pan	080 General 5
Sw 1		011 Xpression	081 General 6
Sw 2		012 Effect 1	082 General 7
Sw 1 & 2		013 Effect 2	083 General 8
Mono Lvl		014 Ctl 14	084 Porta Ctl
Left Lvl		015 Ctl 15	085 Ctl 85
Right Lvl		016 General 1	...
Footpedal		017 General 2	090 Ctl 90
Foot Sw 1		018 General 3	091 FX1 Depth
Foot Sw 2		019 General 4	092 FX2 Depth
ADJUST		020 Ctl 20	093 FX3 Depth
Tempo		...	094 FX4 Depth
On		031 Ctl 31	095 FX5 Depth
Off		(PCM 81 interprets 032 as Bank Select)	096 Data Inc
		033 Ctl 33	097 Data Dec
		...	098 NRPN LSB
		063 Ctl 63	099 NRPN MSB
		064 Sustain	100 RPN LSB
		065 Porta On	101 RPN MSB
		066 Sostenuto	102 Ctl 102
		067 SoftPedal	...
		068 Legato	119 Ctl 119
		069 Hold 2	
MIDI			
P Bend			
A Touch			
Velocity			
Last Note			
Low Note			
High Note			
Clk Comnds			

Pitch The Resonant Chord algorithms: Res1>Plate and Res 2>Plate each have a Pitch row that contains the parameters for setting and controlling the tuning of the effect's resonators. Each set of parameters is presented separately here.

Res 1>Plate Pitch parameters

The voice resonators take the audio impulse from a delay voice and “resonate” it at a desired pitch. Until a resonator is assigned a pitch, it does not resonate. The Res1 effect is a “round robin” pitch assign in that pitch changes are individually assigned to one of (as many as) six voice resonators. For example, assume that all six voice resonators are active in the newly loaded effect, and the first pitch assigned is A4. Voice 1's resonator will be set to the A4 frequency. Assign B4, and Voice 2's resonator will be set to that pitch. In this example, if we assign four more pitches (D5, G4, A#:Bb4, C5, for example), they will be assigned to the Voice 3, 4, 5, and 6 resonators, respectively. If a seventh pitch is assigned, for example, F3, Voice 1's resonator will be changed from A4 to F3 (the other resonators remain at the last assigned pitches).

Assign

This parameter has a range of C0 to G10 (128 pitches) to be convenient for MIDI patching. As the value of this parameter is changed, each “next value” is assigned to the next active pitch resonator of the six possible active resonators. An example might be driving this parameter with a slow LFO to achieve an effect something like strumming the strings of an open piano.

Tuning

This parameter allows you to tune the A440 reference of the box anywhere from 430.0-450.0Hz in 0.1Hz increments.

Active

This parameter allows you to define the number of active voice resonators. You can choose from 0 (no resonators active) to 6. Setting the value to 1 will limit pitch assignments to Voice 1's resonator. Setting the value to 2 will limit the round robin pitch assign to Voices 1 and 2. A value of 6 uses all the voice resonators.

A handy feature of this parameter is the ability to mute resonators. For example, you can patch this parameter to a footswitch or pedal, allowing you to drive the parameter value down to 0 at any time, causing the resonators to stop playing their last assigned pitches. You can then immediately drive it back up to the desired number of active resonators. This “clears” the pitch assign for the resonators, so they don't go back to playing their previous pitches, but rather wait for the next pitch assign to come along before resonating.

Unison

This is a simple On/Off control. When set to On, pitch assigns will be sent simultaneously to all active resonators. When set to Off, pitch assigns will operate in the round robin method described earlier. This allows you to set multiple delay times for all Voices and allows a single pitch assign (rather than 6 individual assigns) to set all the resonators to the same pitch. The pitch will ring out at 6 different delay times from the input impulse.

Res 2>Plate Pitch parameters

The voice resonators take the audio impulse from a delay voice and “resonate” it at a desired pitch. The Res2 effect is a “interval harmonization” pitch assign in that pitch changes generate interval pitches to be assigned to the six voice resonators.

Pitch cont'd.

Position 0 contains all of the pitch parameters which affect the voices in the remaining position in the row. Press **Load/★** to cycle through the selections at position 0: **Assign, Tuning, Active, Key, Scale, Root** and **Rule**.

- Assign** This parameter has a range of C0 to G10 (128 pitches to be convenient for MIDI patching). As the value of this parameter is changed, each voice resonator is set to a pitch at the defined voice's interval (described later).

- Tuning** This parameter allows you to tune the A440 reference of the box anywhere from 430.0-450.0Hz in 0.1Hz increments.

- Active** This parameter allows you to define the number of voice resonators which are active. You can choose from 0 (no resonators active) to 6. Setting the value to 1 will permit only Voice 1's resonator to ring out. Setting the value to 2 will permit Voices 1 and 2 to ring out. A value of 6 uses all the voice resonators. A handy feature of this parameter is the ability to mute resonators. For example, you can patch this parameter to a footswitch or pedal, allowing you to drive the parameter value down to 0 at any time, causing the resonators to stop playing their last assigned pitches. You can then immediately drive it back up to the desired number of active resonators. This “clears” the note assign for the resonators, so they don't go back to playing their previous pitches, but rather wait for the next pitch assign to come along before resonating.

- Key** This parameter establishes a key signature for the pitch selection of the voice intervals. The range of this value is the 12 keys from C, C#, etc., up to B.

- Scale** This parameter establishes one of two scales of semitones: a standard Major scale (intervals of 2, 2, 1, 2, 2, 2, 1 half steps for the scale), or a Harmonic scale (intervals of 2, 1, 2, 2, 1, 3, 1).

- Root** This parameter (with values of 1-7) establishes the root of the scale which, when combined with the scale, defines the mode. For example, with a Major scale selected, roots 1-7 define the modes — Ionian (Major), Dorian, Phrygian, Lydian, Mixolydian, Aeolian (minor) and Locrian.

Pitch cont'd.

Rule This parameter has four values: Round Down, Round Up, Shift Down, and Shift Up. Its exclusive purpose is to tell the interval harmonizer what to do with out-of-key pitch assignments. The values instruct the interval harmonizer as follows:

Round Down Take the input pitch assign, round it down a half-step to an in key pitch, then calculate the interval pitch.

Round Up Take the input pitch assign, round it up a half-step to an in key pitch, then calculate the interval pitch.

Both of these generate in-key harmonization.

Shift Down Take the input pitch assign, round it up a half-step to an in-key pitch, calculate the interval pitch then shift the interval pitch down a half-step.

Shift Up Take the input pitch assign, round it down a half-step to an in-key pitch, calculate the interval pitch then shift the interval pitch up a half-step.

Both of these generate out-of-key harmonization.

Unison and Octave interval pitches strictly follow the input pitch assign. An example of how to use these follows the explanation of Voice Pitch intervals.

V1-V6 Pitch

The Res2>Plate effect has an individual pitch interval control for each voice resonator. The range of each control is up or down any scale interval through five octaves and Unison, for a total of 81 possible settings. For example: an interval of up one-fifth reads "+5th", and an interval of up one octave and a fifth reads "+1 Oct +5th". Down one-fifth reads "-5th", down one octave and a fifth reads "-1 Oct -5th".

An example application of Key, Scale, Root, Rule and Voice Pitch Intervals. Pitch cont'd.

Active resonators set to 3.

The Key is C.

The Scale is Major.

The Root is 1. (C Major Ionian)

Voice 1 Pitch is assigned to Unison.

Voice 2 Pitch is assigned to +3rd.

Voice 3 Pitch is assigned to +5th. (Basic triad)

In-key pitch assignment examples:

1. Pitch Assign: C4. Voice 1 will resonate at C4, Voice 2 at E4, and Voice 3 at G4. (Standard chord in the key based on the tonic)
2. Pitch Assign: D4. Voice 1 will resonate at D4, Voice 2 at F4, and Voice 3 at A4. (Standard chord in the key based on the supertonic)
3. Pitch Assign: E4. Voice 1 will resonate at E4, Voice 2 at G4, and Voice 3 at B4. (Standard chord in the key based on the median)
4. Pitch Assign: B4. Voice 1 will resonate at B4, Voice 2 at D5, and Voice 3 at F5. (Standard chord in the key based on the leading tone)

Out-of-key pitch assignment — Pitch Assign: C#/Db4.

Rule: Round Down: Voice 1 will resonate at C#/Db4, Voice 2 at E4, and Voice 3 at G4 diminished

Round Up: Voice 1 will resonate at C#/Db4, Voice 2 at F4, and Voice 3 at A4 augmented

Shift Down: Voice 1 will resonate at C#/Db4, Voice 2 at E4, and Voice 3 at G#:Ab4 minor

Shift Up: Voice 1 will resonate at C#/Db4, Voice 2 at F4, and Voice 3 at G#:Ab4 Major

Resonance In the Resonant Chord algorithms: Res1>Plate and Res2>Plate, this row contains high cut filter and resonance controls for each voice, as well as master high cut and resonance controls for all voices.

Mstr Res,Mstr HC

Position 0 of the Resonance row contains two master parameters for the six resonator voices: Mstr Res and Mstr HC. (Press **Load/★** to toggle between them.) Mstr Res is a master resonance control for all of the resonator voices. It allows the resonance settings for all voices to be scaled over a range of 0-100%. Use care when adjusting this value. The resonators will put out high levels if the input source has any components that are close in pitch to the resonator tunings.

The Mstr HC control allows the high cut filter settings for all voices to be scaled over a range of 0-100%.

V1Res,V1 HiCut-V6 Res,V6 HiCut

The V1-V6 Res controls set the amount and polarity of feedback for each voice. The HiCut controls set the high frequency cutoff frequency of the 6dB per octave low pass filter for each voice.

RvbDesign The Reverb Design row, available in every algorithm, contains parameters that affect the structural aspects of the reverb effect.

Size

Size sets the rate of build-up of diffusion after the initial period (which is controlled by Diffusion). The Size control changes a reverb sound from very large to very small. Generally, you should set this control to approximate the size of the acoustic space you are trying to create, before adjusting anything else. The size in meters is roughly equal to the longest dimension of the space. Audio is temporarily muted when Size is changed.

Duration

In the Inverse algorithm, Duration determines the length of time, in milliseconds, which passes before the cutoff in Inverse effects.

Attack

Attack is provided in the Plate algorithm to set the sharpness of the initial response to an input signal. High settings cause an explosive sound, while low settings cause the sound to build up more slowly with time. Attack only affects the level of sound within the first 50 milliseconds.

Diffusion

A Diffusion control is provided in all algorithms. It controls the degree to which initial echo density increases over time. High settings of Diffusion result in initial build-up of echo density, and low settings cause low initial build-up. Echo density is also affected by Size; smaller spaces will sound denser. To enhance percussion, use high settings of Diffusion. For clearer and more natural vocals, mixes, and piano music, use low or moderate settings of Diffusion.

Shape, Spread

In the Chamber and Infinite algorithms, Shape and Spread work together to control the overall ambience of the reverberation created by the PCM 81. Shape determines the contour of the reverberation envelope. With Shape all the way down, reverberation builds explosively, and decays quickly. As Shape is advanced, reverberation builds up more slowly and sustains for the time set by Spread. With Shape in the middle, the build-up and sustain of the reverberation envelope emulates a large concert hall (assuming that Spread is at least halfway up, and that Size is suitably large — 30 meters or larger). Low Spread settings result in a rapid onset of reverberation at the beginning of the envelope, with little or no sustain. Higher settings spread out both the buildup and sustain. In the Inverse algorithm, Spread is fixed, and only a Shape control is available.

Rvb Design cont'd.

Def

Available in the Concert Hall and Glide>Hall algorithms, Definition affects the echo density buildup rate during the latter part of the decay period. At 0, the rate is determined by the program material. Raising Definition causes the sound to become choppy — the decrease in density of the echoes creates increasingly distinct, repetitive echo trails.

Depth

Available in the Concert Hall and Glide>Hall algorithms, this control sets the output amplitude envelope, changing the listener's perspective from the front to the rear of the hall.

Spin

Spin affects the movement of the reverberation tail. The object of Spin is to continuously alter the timbre of the reverberant sound. This makes the result more natural, without making the position of instruments unstable. Spin should typically be set to values between 10 and 50. Higher values may make the timbre of piano or guitar unstable.

Chorus

In the Concert Hall and the Glide>Hall algorithms Chorus randomizes delay times and introduces modulation to make reverberation sound less metallic. Increasing Chorus increases the rate of modulation. Because Chorus can cause pitch variation, this parameter should be set with care when using sources with very little pitch wobble (such as guitar or piano). A good practice is to increase the setting until pitch wobble becomes noticeable, then lower it slightly.

Link

This control is available in all algorithms except Inverse. When Link is set to On, the reverb time (Mid Rt) and Spread scale linearly as the Size control is varied. For some special effects, Mid Rt, Spread and Size can be unlinked.

Rvb Width

In all of the 4-Voice algorithms, this control performs the same function as FX Width, but controls the audio of only the reverb and post delay reverb.

Rvb In/Out

Rvb In and Rvb Out control the level of the audio going into and coming out of the reverb effect. Rvb In has a range of Full (0dB) down to -85dB, and Off. Rvb Out has a range of Full (0dB) down to -24dB, and Off. Rvb Out does not affect the level of the pre-echoes set by the Ref Lvl and Eko Fbk parameters.

Rvb Time The Reverb Time row, available in every algorithm, contains parameters that affect the time-based aspects of the reverb effect.

Mid Rt and Low Rt

Mid Rt sets the reverb time for mid-frequency signals. Because low frequency reverb time (Low Rt) is a multiplier of Mid Rt, Mid Rt acts as a master control for the reverb time.

Low Rt sets the reverb time for low-frequency signals, as a multiplier of the Mid Rt parameter. For example, if Low Rt is set to 2X, and Mid Rt is set to two seconds, the low frequency reverb time will be four seconds. For a natural-sounding hall ambience, we recommend values of 1.5X or less.

Low Slope and Mid Slope

These parameters are only available in the Inverse algorithm. Low Slope determines the shape of the reverb envelope for low frequencies. When set to 0, the level of low reverb remains unchanged over its Duration, then cuts off abruptly (depending on the amount of diffusion in use). Setting Low Slope above 0 causes the level of low-frequency reverb to rise smoothly from soft to loud until the sound is cut off. The greater the slope, the softer the initial reverberation and the more pronounced its rise. With negative values, the low frequency reverb drops from its initial level to a quieter one before cutoff. The lower the slope, the more pronounced the dropoff.

Mid Slope is similar to Low Slope, but applies to middle and high frequencies. The actual frequencies affected are determined by Crossover.

Crossover

Crossover sets the frequency at which the transition from Mid Rt to Low Rt takes place. This control should be set at least two octaves higher than the low frequency you want to boost. For example, to boost a signal at 100Hz, set Crossover to 400Hz. (This setting works well for classical music.) Crossover works best around 400Hz for boosting low frequencies, and around 1.5 kHz for cutting low frequencies.

Rt HC

Rt HC sets the frequency above which a 6dB/octave low-pass filter attenuates the reverberated signal. It does not attenuate the reflections. High frequencies are often rolled off with this parameter, resulting in more natural-sounding reverberation.

Pre Delay

Pre Delay adjusts an additional time delay between the input of signal and the onset of reverberation. The maximum range is 930ms. This control is not intended to mimic the time delays in natural spaces. In real rooms, the build-up of reverberation is gradual, and the initial time gap is usually relatively short. Natural spaces are best emulated by adjusting Spread for the desired effective predelay.

Infinite

In the Infinite algorithm, this control is provided to turn the infinite effect On or Off.

RefLvl L&R, RefDly L&R

These controls provide pre-echoes to the left and right channels. The maximum range for Inverse effects is 800ms. In all other effects it is 1.2 seconds. Press **Load/★** to cycle through the selections.

EkoFbk L&R,EkoDly L&R

Available in the Plate, Chamber and Infinite algorithms, these controls provide a pre-echo of 1.2 seconds maximum to the left and right channels, with feedback. Press **Load/★** to cycle through the selections.

PstMix, PstGld

In the 4-Voice Algorithms, these controls adjust a pair of delays after the reverb. Press **Load/★** to cycle through the selections: **PstMix**, **Pst Dly L&R**, **GldResp** and **GldRange**.

Pst Mix

In the 4-Voice Algorithms, this parameter controls the mix of the reverb audio with the post delayed reverb audio (0-100%).

Pst Dly L&R

In the 4-Voice Algorithms, these parameters control the time of the post delay. The available range of 0-682ms can produce a variety of pitch shift, flange, or stutter effects on the reverb audio. (Expanding PCM 81 memory will extend this range to 1365ms.)

GldResp

In the 4-Voice Algorithms, this parameter controls the response of the glide of the post delay. The range is 0-100, with a typical default setting of 50. This setting provides a good, real-time glide. Changing a delay time from 0 to 100 ms with a glide response of 0, causes the glide delays to take a minute or more, allowing subtle changes in delay times and echoes with no noticeable pitch shifting. A glide response of 100 is ultrafast, causing high speed pitch shift, and even chirping effects with large delay time changes.

GldRange

In the 4-Voice Algorithms, this parameter controls the range over which changes to the post delay time will be performed as a glide. For example, if **GldRange** is set to 100ms, and the **PstDly L** and **R** times are varied over a range of 100ms, those delays will glide smoothly from one delay time to another. Suddenly changing the delay time to a greater value, such as 200ms, will cause the audio delay to change instantly to the new time without gliding. The available range is 0-682ms, with 0 causing all delay time changes to be "instantaneous", and 682 causing all delay time changes to glide.

Rvb Time cont'd.

You can set and display delay values in units of time, or with tempo values. Press **Up** and **Tempo** simultaneously to toggle between these two options. When time units are selected, delay times are set and displayed in milliseconds (From 0ms to the maximum available delay for that parameter). When tempo values are selected, delay values are set and displayed as a ratio of echoes to beats (from 24:1 to 1:24). This will automatically synchronize the delay to the current tempo (MIDI, Internal, or TAP — see *Tempo Mode* in Chapter 2.) For example, a delay setting of 1:2 (1 echo for every 2 beats) will produce half-note delay rhythms synchronized to the current tempo.

The Pitch Algorithms

About the Pitch Algorithms

The PCM 81 contains 7 Pitch algorithms, each combined with uncompromised Lexicon reverb, to provide a wide range of useful pitch shift effects, ranging from pitch correction to unique special effects.

A Quad>Hall algorithm provides a 4-voice pitch shifter, combined with the PCM 81 Concert Hall reverb. The Dual-Chmb, Dual-Plt, Dual-Inv, Stereo-Chmb and VSO-Chmb algorithms use the PCM 81 Dual FX structure for unparalleled flexibility and ease of use. The Pitch Correct algorithm provides tools specifically designed for correction of previously recorded vocal tracks.

The five Dual FX-type algorithms: Dual-Chmb, Dual-Plt, Dual-Inv, Stereo-Chmb and VSO-Chmb, each contain two independent effect blocks — one of three stereo reverbs and one of three different stereo pitch shift blocks. Each algorithm also has the full set of modulation and patch features of the PCM 81 4-Voice and 6-Voice algorithms.

Configuration and routing control is provided by a dedicated Submixer incorporated into each of these five algorithms. Any combination of reverb and effects blocks can be arranged in any configuration simply by changing the Submixer's controls. As the Submixer has continuously variable controls, you can alter the configurations on the fly — changing, for example, from reverb-into-effects to effects-into-reverb. (Use of the Submixer, which appears as a separate row on the PCM 81 parameter matrix, is described fully after the algorithm and parameter descriptions.) Parameters which are common to the 4-Voice and 6-Voice algorithms are not repeated here.

The Pitch Correct algorithm, designed for use with monophonic vocal sources, is described at the end of this section.

About Pitch Shifting

Digital pitch shifting requires temporal compression or expansion of audio data and the addition or removal of small audio segments to compensate.

Raising pitch is accomplished by compressing the audio data — making it run faster. As this shortens the audio segment, a copy of a section of the audio being processed is added to fill out the remaining time. To lower the pitch, the signal is expanded, requiring the removal of a section of the audio. These operations are performed repeatedly at high speed, at intervals which vary according to the type of audio signal being processed.

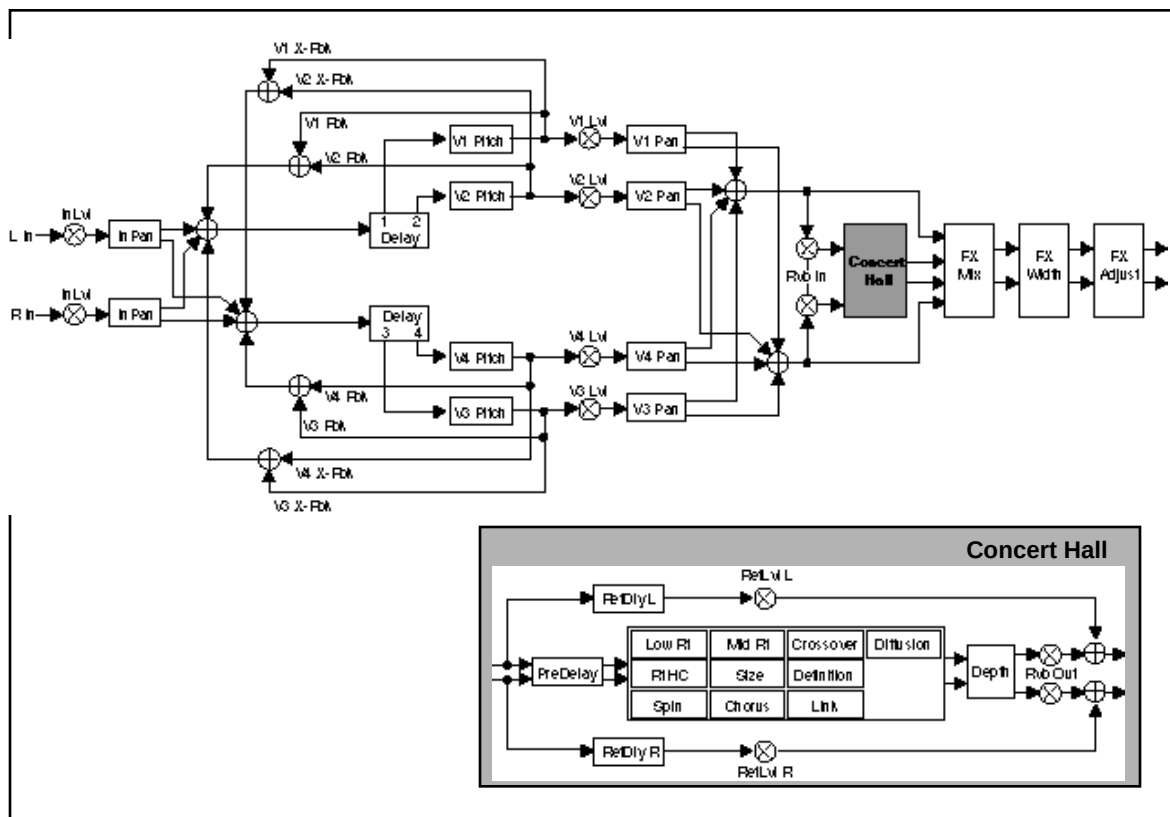
At each point where audio is added or removed, the audio signal must be rejoined, or spliced. If added or removed segments were perfectly correlated with the remaining audio, splicing would be seamless. In reality, these splice points are the source of audible artifacts which can be minimized, but not completely eliminated.

Much of the art of pitch shifting, therefore, is in determining the appropriate intervals for sampling and the optimal points for splicing. As the PCM 81 Pitch algorithms gives you significant control over these operations, you may find it helpful to keep in mind some of the other factors which can affect pitch shift quality:

- **Amount of pitch shifting**
Large shift intervals require more splicing. As each splice can be the source of an artifact, increasing their number increases the likelihood of audible artifacts.
- **Pitch-complexity of the input material**
Pure tones have simple waveforms which can be correlated easily for smooth splicing. Pitch shifting of simple tones such as those generated by a solo voice or a monophonic instrument generally produces fewer artifacts than pitch shifting of wide bandwidth material such as piano chords, full orchestra, etc.
- **Low frequency content of the input material**
Pitch shifting low frequency material requires the addition or removal of larger segments of audio than high frequency material. The increased size of these segments can introduce delays which are heard as artifacts.
- **Transient content of the input material**
When large amounts of pitch shifting are applied to complex input material with dense percussive transients (drums, strummed mandolin, complex speech, etc.) some of the transients may lose definition or be shifted slightly in time.

Quad>Hall The Quad>Hall algorithm, is a 4-voice pitch shifter in series with the Concert Hall reverb. As in the multi-voice algorithms, each voice has its own delay, level, feedback, cross-feedback and panning controls. The maximum delay time for each voice is 1.250 seconds.

In this algorithm, the reverb effect is fixed in position following the pitch shifters, with a final Mix control allowing control over the amount of reverb in the processed sound. Quad>Hall is designed for creating pitch effects, and is an obvious choice for creating sequences of notes or chords, melody fragments, or small amounts of pitch shift for doubling (or quintupling) effects.

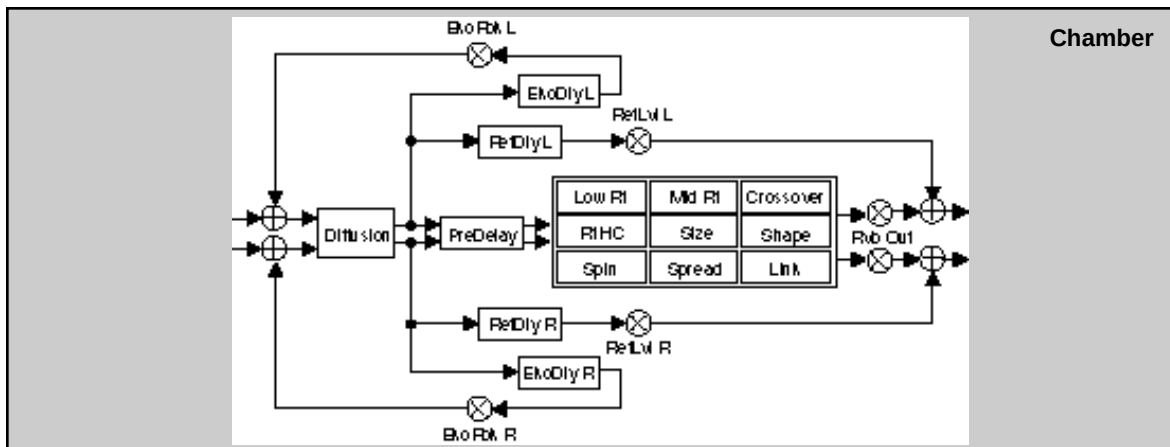
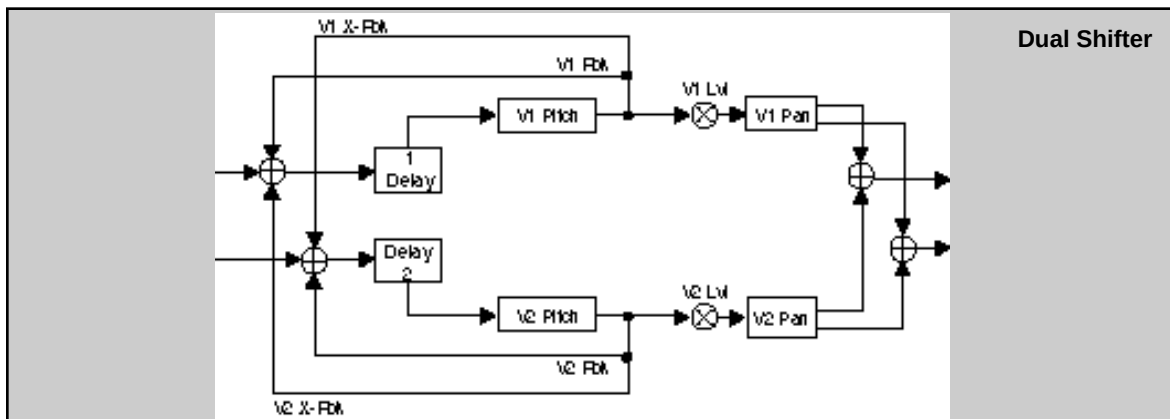


The PCM 81 Concert Hall provides very clean reverberation. It is designed to remain behind the direct sound — adding ambience, but leaving the source unchanged. This effect has a relatively low initial echo density which builds up gradually over time.

0	Control+	0.0 Min	0.1 LvlAdj	0.2 *RelLvlPan	0.3 BK Min	0.4 BK Midh	0.5 *RelLvlST				
1	Pwb Time	1.0 LowR	1.1 Mid R	1.2 Crossover	1.3 R HC	1.4 Pre Dela ₂	1.5 *RelLvlDL				
2	Pwb Design	2.0 6db	2.1 Diffusion	2.2 Del	2.3 Depth	2.4 6ph	2.5 Chorus	2.6 Lnk	2.7 *Rub h Dur		
3	Level+	3.0 Master	3.1 Voice1	3.2 Voice2	3.3 Voice3	3.4 Voice4					
4	Delay Time	4.0 Master	4.1 Voice1	4.2 Voice2	4.3 Voice3	4.4 Voice4					
5	Pitch	5.0 *MinCent+	5.1 Voice1	5.2 Voice2	5.3 Voice3	5.4 Voice4					
6	Feedback	6.0 *Min Rbk	6.1 *V1 Rbk	6.2 *V2 Rbk	6.3 *V3 Rbk	6.4 *V4 Rbk					
7	Panning	7.0 *MinMidh	7.1 Voice1	7.2 Voice2	7.3 Voice3	7.4 Voice4					
8	Modulation	8.0 *ModLFO	8.1 *ModRR Env	8.2 Mod EnvL	8.3 Mod EnvR	8.4 *ModLach	8.5 *ModEnvL	8.6 *ModEnvC			
Patch++	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7	Patch 8	Patch 9	

The following five Pitch FX algorithms: **Dual-Chmb**, **Dual-Plt**, **Dual-Inv**, **Stereo-Chmb** and **VSO-Chmb** are created from 3 Reverb blocks and 3 Pitch Shift Effect blocks. The individual reverb and effect blocks are described below, followed by the diagrams and matrixes and parameter descriptions for the algorithms. Descriptions of the controls common to all of these algorithms is presented in following section *Using the Submixer*.

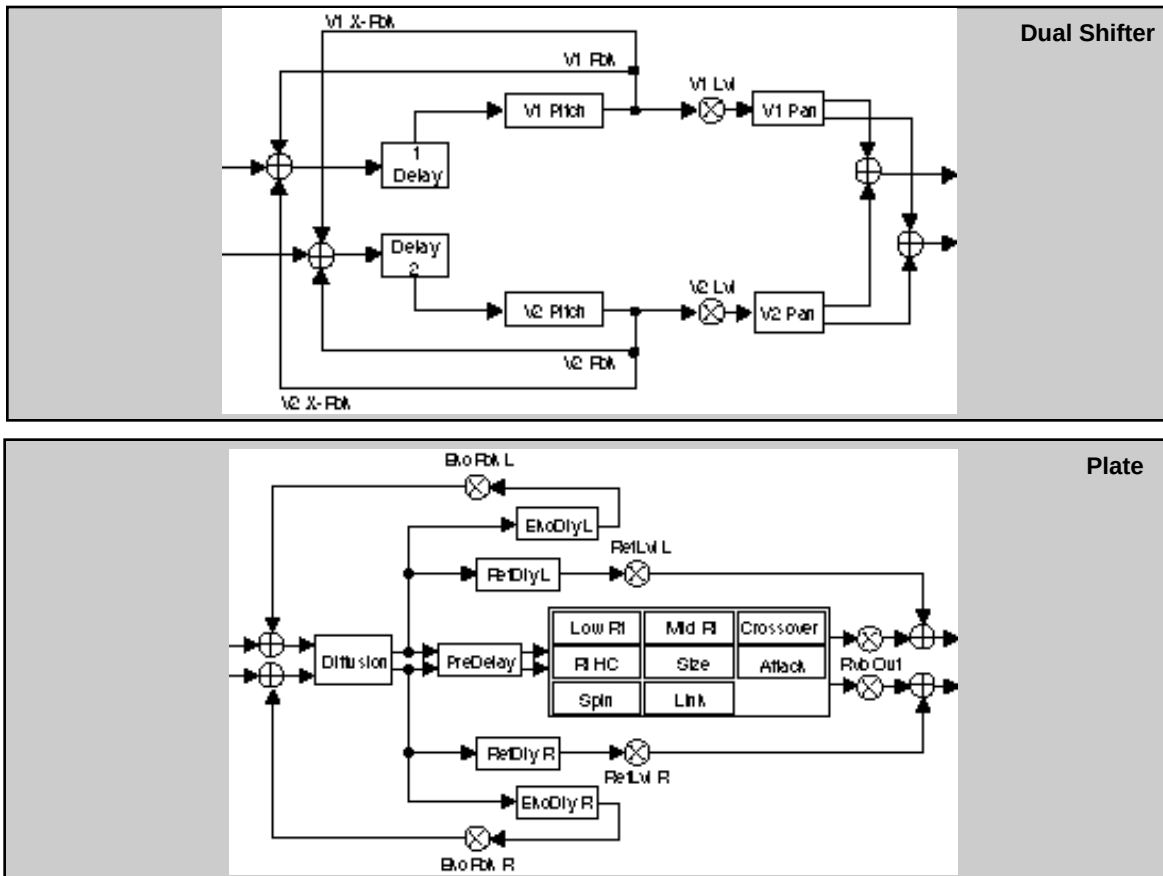
Dual-Chmb This algorithm includes a dual pitch shifter combined with the Chamber reverb. The pitch shifter has two voices. Each voice has independent controls for pitch, level, delay, pan, feedback and cross-feedback. The pitch of each voice is adjustable up or down over a 3 octave range with a resolution of 1 cent. Maximum delay time for each voice is 1.250 seconds.



The Chamber reverb produces an even, relatively dimensionless reverberation, with little change in color as the sound decays. The initial diffusion is similar to the Concert Hall algorithm, but the sense of space and size is much less obvious. This characteristic, along with the low color in the decay tail, makes Chamber useful on a wide range of material. It is especially useful on spoken voice, giving a noticeable increase in loudness with very low coloration.

0 Control+	0.0 Mix	0.1 LvlAdj	0.2 %InLvlPan	0.3 %R Width	0.4 %DJUST					
1 Submixer	1.0 %ends	1.1 %Reums	1.2 %Routing	1.3 %Mix	1.4 %InLvl	1.5 %In W	1.6 %HiCut	1.7 %LoCut	1.8 %OutLvl	1.9 %Out W
2 PubTime	2.0 %LowR	2.1 %Mid R	2.2 %Crossover	2.3 %R HC	2.4 %Pre Dela.	2.5 %RelLvlMDL	2.6 %RelRkMDL			
3 PubDesign	3.0 %Dze	3.1 %Diffusion	3.2 %Shape	3.3 %spread	3.4 %pH	3.5 %Link	3.6 %Pub Out			
4 Level+	4.0 %Master	4.1 %Voice1	4.2 %Voice2							
5 DelayTime	5.0 %Master	5.1 %Voice1	5.2 %Voice2							
6 Pitch	6.0 %MasterCents	6.1 %Voice1	6.2 %Voice2							
7 Feedback	7.0 %MR Rk	7.1 %VR Rk	7.2 %VC Rk							
8 Panning	8.0 %MRWidth	8.1 %Voice1	8.2 %Voice2							
9 Modulation	9.0 %ModLFO	9.1 %ModLFR Env	9.2 %ModEnvL	9.3 %ModEnvR	9.4 %ModLch	9.5 %ModSml	9.6 %ModSml2			
Patch++	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7	Patch 8	Patch 9

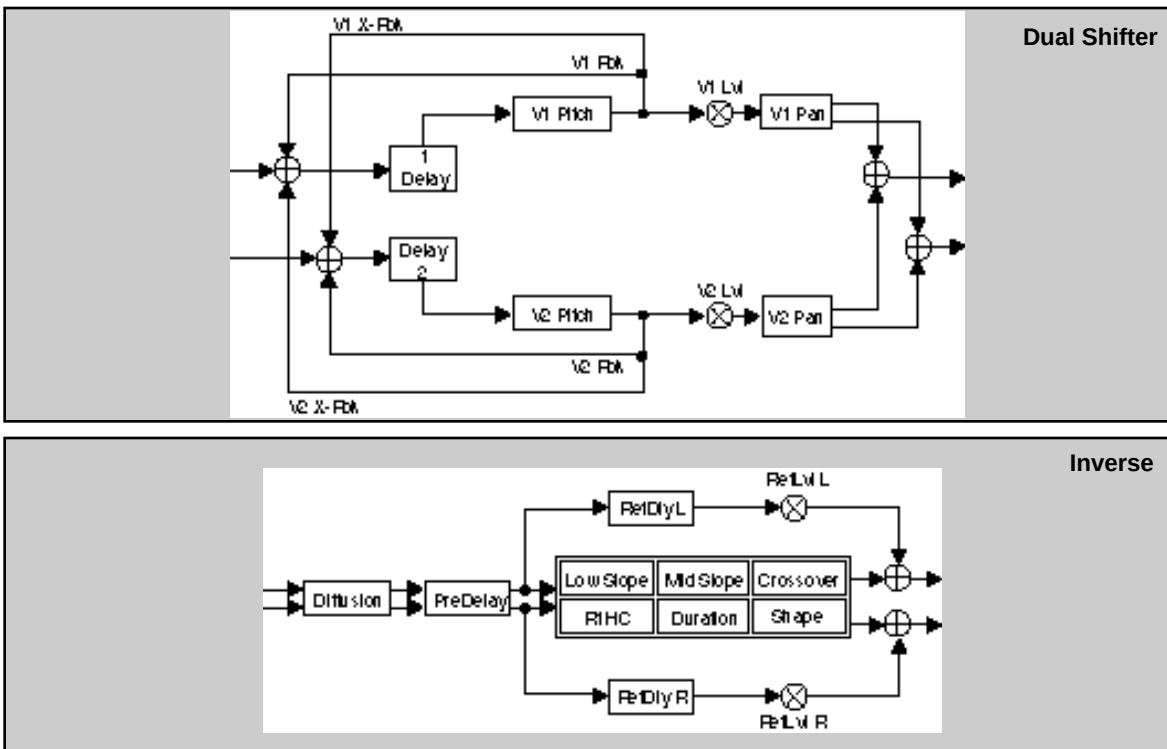
Dual-Pit This algorithm includes a dual pitch shifter combined with the Plate reverb. The pitch shifter has two voices. Each voice has independent controls for pitch, level, delay, pan, feedback and cross-feedback. The pitch of each voice is adjustable up or down over a 3 octave range with a resolution of 1 cent. Maximum delay time for each voice is 1.250 seconds.



The Plate reverb mimics the sound of metal plates, with high initial diffusion and a relatively bright sound, making it a good choice for enhancing any type of percussion.

0	Control+	0.0	Mix	0.1	LutAdjW	0.2	*InLutPan	0.3	RC Width	0.4	*PDJUS7																				
1	Submixer	1.0	Sends	1.1	Reverb	1.2	Routing	1.3	*MixA	1.4	*InLut	1.5	*In IN	1.6	*HiCut	1.7	*LoCut	1.8	*OutLut	1.9	*Out IN										
2	PubTime	2.0	LowR	2.1	Mid R	2.2	Crossover	2.3	RC RC	2.4	Pre Dela.	2.5	*PreLutDL	2.6	*Bto RnDL																
3	PubDesign	3.0	Size	3.1	Diffusion	3.2	Attack	3.3	6pin	3.4	Link	3.5	Pub Out																		
4	Level+	4.0	Master	4.1	Voice1	4.2	Voice2																								
5	DelayTime	5.0	*Master	5.1	Voice1	5.2	Voice2																								
6	Pitch	6.0	*MnxCent+	6.1	Voice1	6.2	Voice2																								
7	Feedback	7.0	*MnR Rbk	7.1	*V1 Rbk	7.2	*V2 Rbk																								
8	Panning	8.0	MnRWidth	8.1	Voice 1	8.2	Voice 2																								
9	Modulation	9.0	*ModLFO	9.1	*ModLFR Env	9.2	Mod EnvL	9.3	Mod EnvR	9.4	*ModLach	9.5	*ModEnvM1	9.6	*ModEnvM2																
End++		Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7	Patch 8	Patch 9																				

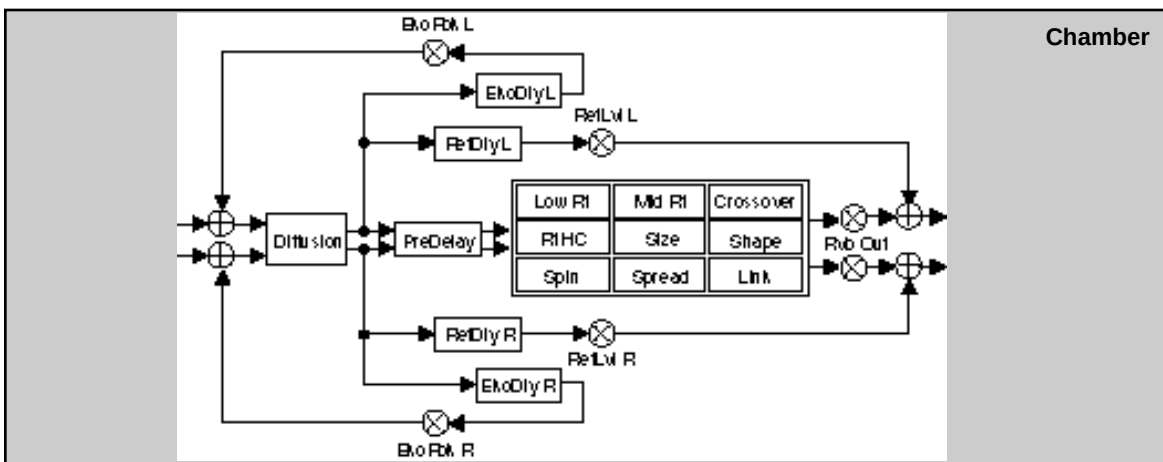
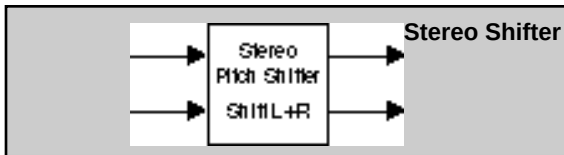
Dual-Inv This algorithm includes a dual pitch shifter combined with the Inverse reverb. The pitch shifter has two voices. Each voice has independent controls for pitch, level, delay, pan, feedback and cross-feedback. The pitch of each voice is adjustable up or down over a 3 octave range with a resolution of 1 cent. Maximum delay time for each voice is 1.250 seconds.



Inverse reverb allows you to vary the slope of the initial portion of the reverb envelope. The slope can decay, remain level, or rise over a variable time interval. When the time interval is up, the reverberation cuts off abruptly. The resulting effect is similar to a gate, but is not at all dependent on the level or complexity of the input signal. Slopes are adjustable over a negative, even, or positive slope. Positive slopes create inverse effects, while more even slopes create gated effects. Negative slope values have rather natural reverb tails.

0 Control+	0.0 Min	0.1 LutAdjw	0.2 *InLutHPan	0.3 RC Midh	0.4 *PDJLUT					
1 Submixer	1.0 Sends	1.1 Returns	1.2 Routing	1.3 *MIn	1.4 *InLut	1.5 *In IN	1.6 *HICut	1.7 *LoCut	1.8 *OutLut	1.9 *Out IN
2 PwbTime	2.0 LowSlope	2.1 Mid Slope	2.2 Crossover	2.3 *FIR HC	2.4 Pre Dela.	2.5 *PwLutDL				
3 PwbDesign	3.0 Duration	3.1 Diffusion	3.2 Shape							
4 Level+	4.0 Master	4.1 Voice1	4.2 Voice2							
5 DelayTime	5.0 *Master	5.1 Voice1	5.2 Voice2							
6 Pitch	6.0 *MwGen+	6.1 Voice1	6.2 Voice2							
7 Feedback	7.0 *MwR FB	7.1 *V1 FB	7.2 *V2 FB							
8 Panning	8.0 *MwMidh	8.1 Voice1	8.2 Voice2							
9 Modulation	9.0 *ModLFO	9.1 *ModLFR Env	9.2 Mod EnvL	9.3 Mod EnvR	9.4 *ModLach	9.5 *ModSawL	9.6 *ModSawC			
Patches+	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7	Patch 8	Patch 9

Stereo-Chmb The Stereo-Chmb algorithm is optimized for the best possible shifted audio quality while maintaining the stereo imagery of the source material. Pitch is adjustable up or down over a 3 octave range with a resolution of 1 cent. This effect is combined with the Chamber reverb.



The Chamber reverb produces an even, relatively dimensionless reverberation, with little change in color as the sound decays. The initial diffusion is similar to the Concert Hall algorithm, but the sense of space and size is much less obvious. This characteristic, along with the low color in the decay tail, makes Chamber useful on a wide range of material. It is especially useful on spoken voice, giving a noticeable increase in loudness with very low coloration.

0 Controls	0.0 Mix	0.1 LvlAdj	0.2 *InLvlPan	0.3 *RcWidth	0.4 *ReDlyT					
1 Submixer	1.0 *ends	1.1 Returns	1.2 Routing	1.3 *Mix	1.4 *InLvl	1.5 *In H	1.6 *RfCur	1.7 *LoCur	1.8 *OutLvl	1.9 *Out H
2 PreTime	2.0 LowRf	2.1 MidRf	2.2 Crossover	2.3 RfHC	2.4 Pre Del.	2.5 *ReLvlDL	2.6 *ReRfMDL			
3 PreDesign	3.0 Size	3.1 Diffusion	3.2 Shape	3.3 Spread	3.4 Spin	3.5 Link	3.6 Rfb Out			
4 Pitch	4.0 *LowPitch	4.1 *ShiftL+R								
5 Modulation	5.0 *ModLFO	5.1 *ModLFR Env	5.2 ModEnvL	5.3 ModEnvR	5.4 *ModLch	5.5 *ModSml	5.6 *ModSml2			
Presets	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7	Patch 8	Patch 9

VSO-Chmb

The VSO-Chmb algorithm provides the same high quality shifter as Stereo-Chmb, designed for on the fly pitch correction of varispeed playback sources. A varispeed parameter allows you to match the varispeed setting of the playback source. The algorithm can match speeds ranging from +55.00% through -35.48% with a resolution of .01%.

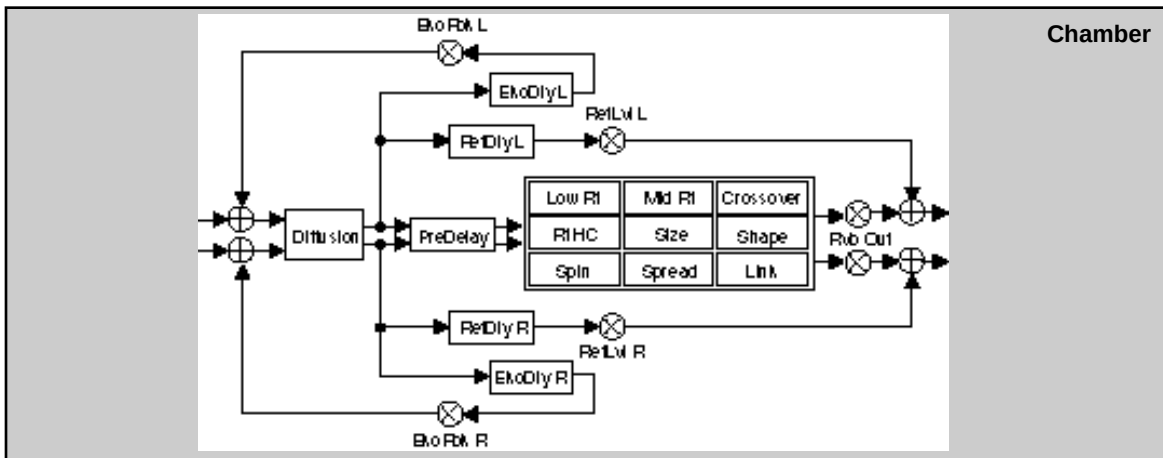
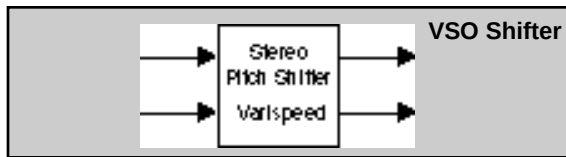
This algorithm is a utility program designed to provide pitch correction of varispeed material. This is a true stereo algorithm which maintains the stereo image of source material. It can be used in applications that require on the fly compression/expansion. Simply match the value of the Varispeed parameter to the varispeed setting of the playback source.

For example, to compress a 30 second spot to 24 seconds, the playback speed must be increased by 20%. The spot now has the correct run length, but increasing the speed created an upward pitch shift of 386 cents. Run the shifted audio through the VSO-Chmb algorithm. Set the Varispeed parameter to +20%. This shifts the pitch down 386 cents. The spot now has a run length of 24 seconds and is at its original pitch.

To expand a 28 second spot to 30 seconds, the playback speed must be decreased by 7.14%. Set both the playback source and the VSO Shifter to -7.14%. The spot will remain at its original pitch and have a run length of 30 seconds.

Like the Stereo-Chmb algorithm, VSO-Chmb is combined with a stereo chamber reverb. This adds considerable flexibility. You can for example, add reverb to the material as it is being pitch corrected (great for voiceovers, etc.).

Note that there is no direct synchronization of the pitch shift with the playback source. The accuracy of the pitch correction is dependent upon the accuracy of the playback source's varispeed mechanism and display.



The Chamber reverb produces an even, relatively dimensionless reverberation, with little change in color as the sound decays. The initial diffusion is similar to the Concert Hall algorithm, but the sense of space and size is much less obvious. This characteristic, along with the low color in the decay tail, makes Chamber useful on a wide range of material. It is especially useful on spoken voice, giving a noticeable increase in loudness with very low coloration.

0 Control+	0.0 Mix	0.1 LowAdj	0.2 RelLvlPan	0.3 Rb Width	0.4 PreDelayT					
1 Submixer	1.0 Sends	1.1 Returns	1.2 Routing	1.3 Mix	1.4 RelLvl	1.5 In Mix	1.6 RI Cut	1.7 Lo Cut	1.8 HLo Cut	1.9 Out Mix
2 Rvb Time	2.0 Low RI	2.1 Mid RI	2.2 Crossover	2.3 RI HC	2.4 Pre Delay	2.5 RelLvl DL	2.6 B.o. Rbk DL			
3 Rvb Design	3.0 Size	3.1 Diffusion	3.2 Shape	3.3 Spread	3.4 Spin	3.5 Link	3.6 Rvb Out			
4 Pitch	4.0 Low Pitch	4.1 Varispeed								
5 Modulation	5.0 ModLFO	5.1 ModLPR Env	5.2 Mod EnvL	5.3 Mod EnvR	5.4 ModL Lch	5.5 ModL Mix 1	5.6 ModL Mix 2			
Presets++	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7	Patch 8	Patch 9

MstrCents

This is a master pitch offset control for all voices. The value dialed in here will be added to all voices. This control allows you to transpose pitch voices while keeping the relative interval(s) between them constant.

MstrScale

This is a master pitch scaling control for all voices. The value of each pitch voice is multiplied by this percentage. This control allows you to shrink or enlarge the relative interval(s) between all voices.

Low Pitch

This parameter allows you to optimize the pitch shifter for the low frequency content of the input material. Use low values when shifting single note material with low frequencies, polyphonic sources or full bandwidth material. Note that the lower this value, the longer the pitch shifted material is delayed. At the lowest setting, the amount of delay is about 42ms. At the highest setting it is about 12ms.

Use this parameter to minimize the amount of delay inherent in the pitch shift process. If the maximum delay of about 40ms is not a problem, you can simply set this value to its lowest setting. To minimize the delay for single note pitch shifting, set it to a value that matches the pitch of the lowest note in the input material. If you are pitch shifting polyphonic input material you may get better results by setting the value lower than the lowest note in the input material.

Lowest note of some typical sources:

Piano	A0
Bass guitar	E1
Guitar	E2
Trombone	F2
Tenor Sax	Bb2
Vocal: bass	Eb2
Vocal: baritone	Ab2
Vocal: tenor	Db3

Splice

This parameter allows you to change the number of milliseconds used to crossfade the splices that are constantly occurring whenever pitch is shifted. For most material a value of 4ms is appropriate. Higher values will produce longer/smooth crossfades, but the intelligibility of transients and high frequency content may be affected. Lower values will reproduce transients more accurately and have more high frequency content, but the splices may be more audible.

GldResp

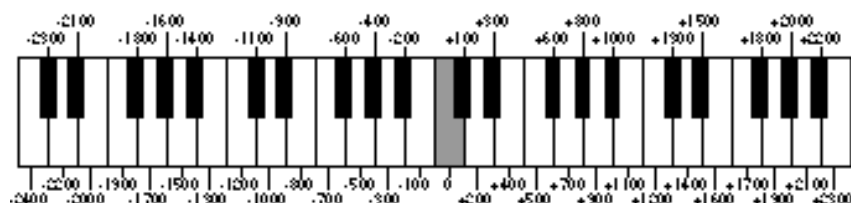
This parameter is similar to the GldResp parameter in the Master Delay parameter of the delay and reverb algorithms. It sets the pitch shifters responsiveness to modulation by internal controllers (LFO, AR, etc.) and MIDI controllers (AfterTouch, Velocity, etc.). It has no effect on the quality of the pitch shifted sound. At the highest setting (100) the pitch will track the modulation source very closely. At lower settings, the pitch will lag behind the modulation source. Use lower settings to smooth out modulation effects.

The Pitch Parameters

Parameter Descriptions cont'd.

Voice 1-4 (or 1-2)

These parameters determine the amount of pitch shift for the left and right input channels. In the Quad Shift algorithm, Voices 1 and 2 are left shifts and 3 and 4 are right shifts. In the 2-Voice algorithms, Voice 1 is left shift and Voice 2 is right shift. The value is given in cents (one cent is equal to 1/100th of a semitone — there are 1200 cents per octave, as shown below). The full range of pitch shift is up or down three octaves (± 3600 cents). The range is adjustable in increments of a single cent. These values can be modulated smoothly with patch sources to create vibrato, Doppler, whammy bar and other pitch-related special effects.



Shift L+R

In the Stereo Shift algorithm, this parameter determines the amount of pitch shift for both the left and right channels. This is a true stereo pitch shifter. The left and right channels are sample synchronous to maintain stereo imaging.

Tips for Setting Pitch Parameters

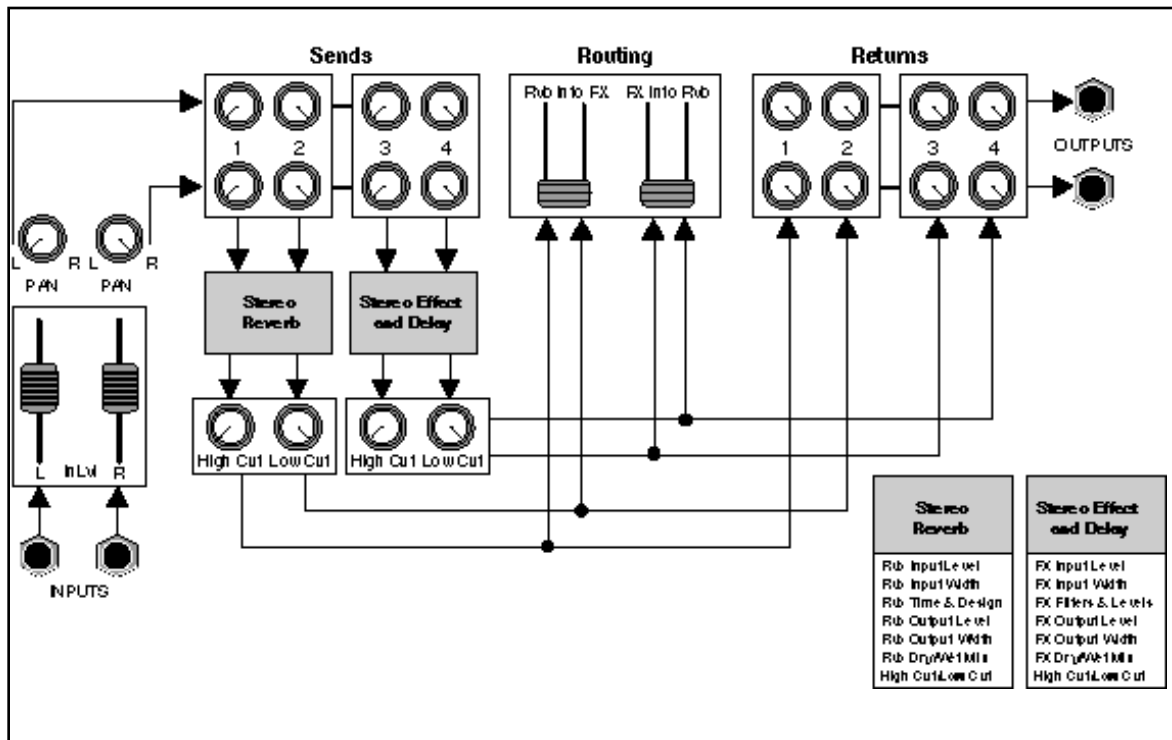
All pitch shifting algorithms will produce some amount of glitches and all pitch shift algorithms have some amount of built in delay. Single note sources shift best. When shifting single notes, Low Pitch should be set at or below the lowest pitch in the input material.

When shifting chords or full bandwidth material, first set the amount of shift. If you need to minimize the delay caused by the shifter, set Low Pitch to the lowest value and slowly increase it to the highest value that doesn't produce unacceptable glitches.

The Splice parameter generally works best for most material when set to 4ms. If the shifted audio sounds a bit harsh, you can smooth it out by raising this value. If there is a loss of intelligibility or rhythmic accuracy in the shifted audio, try setting Splice to a smaller value. As with most effect parameters, you should rely on your ears to find the best settings for a given situation.

Expanded configuration control is available in those Pitch algorithms which have a Submixer incorporated into each algorithm. This Submixer lets you arrange any combination of reverb and effects blocks in any configuration simply by changing the three main Submixer controls: Sends, Returns and Routing. Additional controls are provided for each block: Input Level, Input Width, Output Level, Output Width, High Cut and Low Cut filters, and Wet/Dry Mix.

Using the Submixer



The Dual-Chmb, Dual-Plt, Dual-Inv and Stereo-Chmb algorithms contain a complete dual-channel effects mixer.

The Submixer appears as row 1 on the parameter matrix. The first three parameters in the row are the basic Submixer controls: Sends, Returns, and Routing. Any configuration can be achieved by simply adjusting these three parameters.

1	Submixer	1.0 Sends	1.1 Returns	1.2 Routing	1.3 *Hil	1.4 *HilL	1.5 *HilR	1.6 *HilCut	1.7 *LoCut	1.8 *HilCut	1.9 *HilR
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These parameters provide complete configuration and routing control of the effects.

The key to using the Sub Mixer parameters is understanding what the Sends and Returns parameters do, and how they interact with the Routing Parameters.

Sends The Sends parameter routes the PCM 81's two panned input signals to the four effect inputs (the *Rvb* and *FX* blocks). This parameter has a range of 0–300. The value of this parameter determines what mix of the left and right input signals is fed to the stereo inputs of the *Rvb* and *FX* blocks. The values 0, 100, 150, 200, 300 correspond to specific configurations and are labeled with text instead of numbers.

Sends Value	Rvb Sends	FX Sends
Stereo (0)	L to left Rvb In R to right Rvb In	L to left FX In R to right FX In
L=Rvb, R=FX (100)	L to left Rvb In L to right Rvb In	R to left FX In R to right FX In
Mono (150)	L+R to left Rvb In L+R to right Rvb In	L+R to left FX In L+R to right FX In
L=FX, R=Rvb (200)	R to left Rvb In R to right Rvb In	L to left FX In L to right FX In
Stereo (300)	L to left Rvb In R to right Rvb In	L to left FX In R to right FX In

Returns The Returns parameter routes the four effects outputs to the PCM 81's stereo outputs. This parameter has a range of 0–300. The value of this parameter determines what mix of the stereo *Rvb* and *FX* outputs is fed to the PCM 81's stereo outputs. The values 0, 100, 150, 200, 300 correspond to specific configurations and are labeled with text instead of numbers.

Returns Value	Rvb Returns	FX Returns
Stereo (0)	left Rvb Out to L right Rvb Out to R	left FX Out to L right FX Out to R
Rvb=L, FX=R (100)	left Rvb Out to L right Rvb Out to L	left FX Out to R right FX Out to R
Mono (150)	left + right Rvb Out to L left + right Rvb Out to R	left + right FX Out to L left + right FX Out to R
FX=L, Rvb=R (200)	left Rvb Out to R right Rvb Out to R	left FX Out to L right FX Out to L
Stereo (300)	left Rvb Out to L right Rvb Out to R	left FX Out to L right FX Out to R

The Routing parameter provides continuously variable control of the FX and Rvb effect routing: Rvb and FX in parallel, Rvb into FX (series) and FX into Rvb (reverse series). This parameter has a range of 0–400. The value of this parameter determines the routing/ordering of the stereo Rvb and FX effects. The values 0, 100, 150, 200, 300, 400 correspond to specific configurations and are labeled with text instead of numbers.

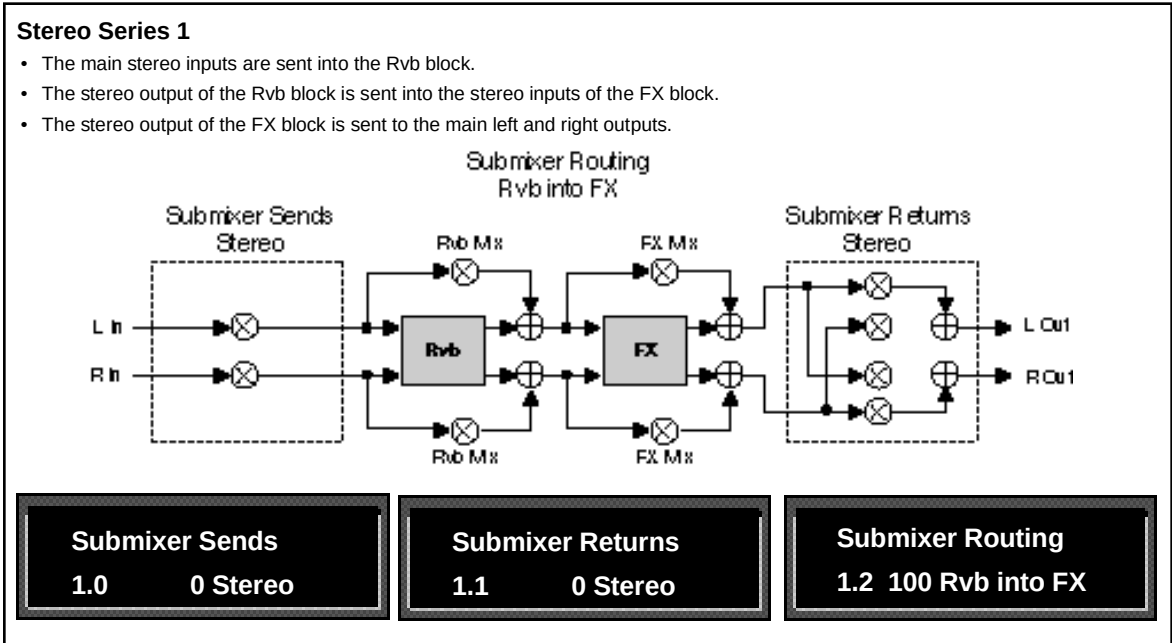
Routing

Routing Value	Effects Routing
Parallel (0)	Rvb and FX are parallel
Rvb into FX (100)	Rvb outputs into FX inputs (series 1)
Parallel (200)	Rvb and FX are parallel
FX into Rvb (300)	FX outputs into Rvb inputs (series 2)
Parallel (400)	Rvb and FX are parallel

Note that the routing value takes precedence over the Sends and Returns values. For example, if Routing is set to Rvb into FX (100), no input signal is routed directly to the FX inputs — and the Rvb outputs aren't sent directly to the PCM 81's main outputs.

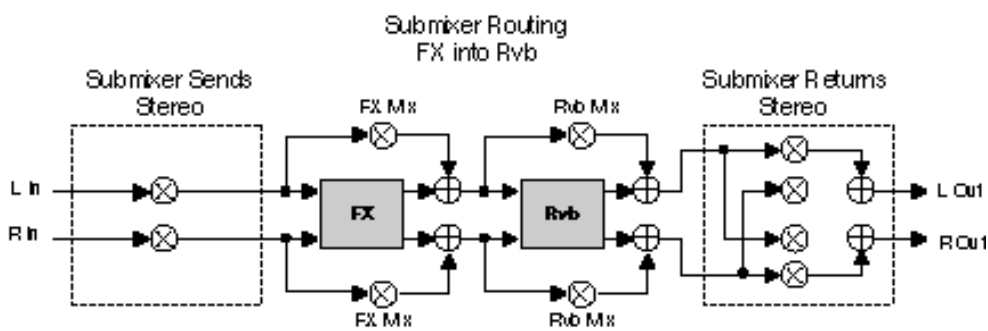
It is possible to arrange the stereo Rvb and FX inputs and outputs into a variety of useful configurations by simply adjusting the Sends, Returns and Routing parameters. The following illustrations show several of these configurations. Each illustration contains a block diagram showing signal flow through the Submixer, as well as the displayed values for the Sends, Returns, and Routing parameters that will produce the configuration.

Useful Configurations

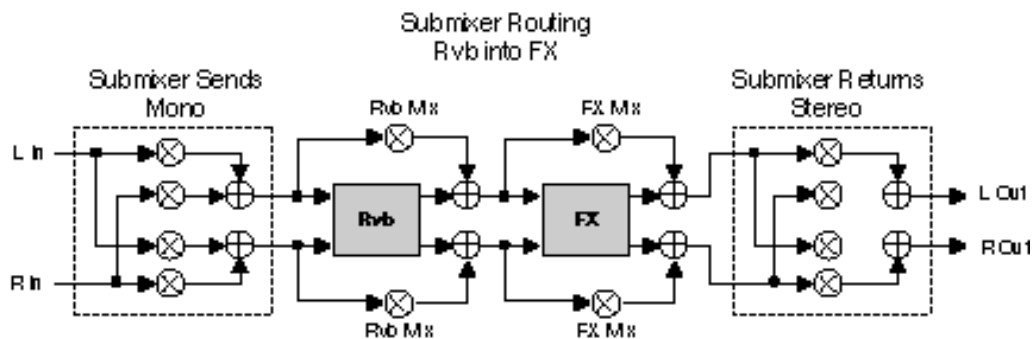


Stereo Series 2

- The main stereo inputs are sent into the FX block.
- The stereo output of the FX block is sent into the stereo inputs of the Rvb block.
- The stereo output of the Rvb block is sent to the main left and right outputs.

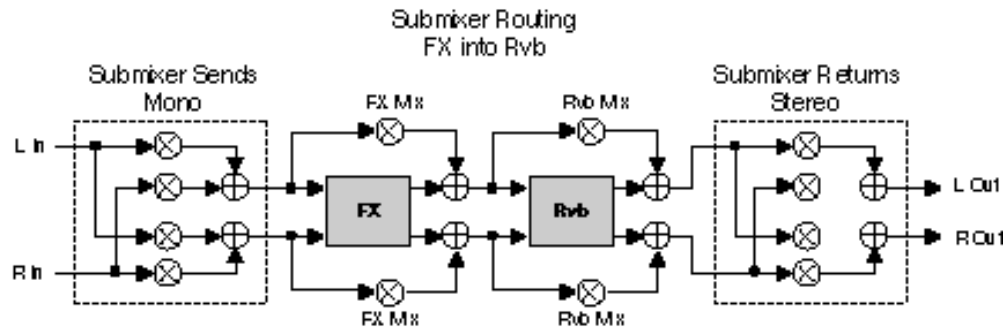
**Submixer Sends****1.0 0 Stereo****Submixer Returns****1.1 0 Stereo****Submixer Routing****1.2 300 FX into Rvb****Mono In/Stereo Out Series 1**

- A mono mix of the left and right inputs is sent into the Rvb block.
- The stereo output of the Rvb block is sent into the stereo inputs of the FX block.
- The stereo output of the FX block is sent to the main left and right outputs.

**Submixer Sends****1.0 150 Mono****Submixer Returns****1.1 0 Stereo****Submixer Routing****1.2 100 Rvb into FX**

Mono In/Stereo Out Series 2

- A mono mix of the left and right inputs is sent into the FX block.
- The stereo output of the FX block is sent into the stereo inputs of the Rvb block.
- The stereo output of the Rvb block is sent to the main left and right outputs.



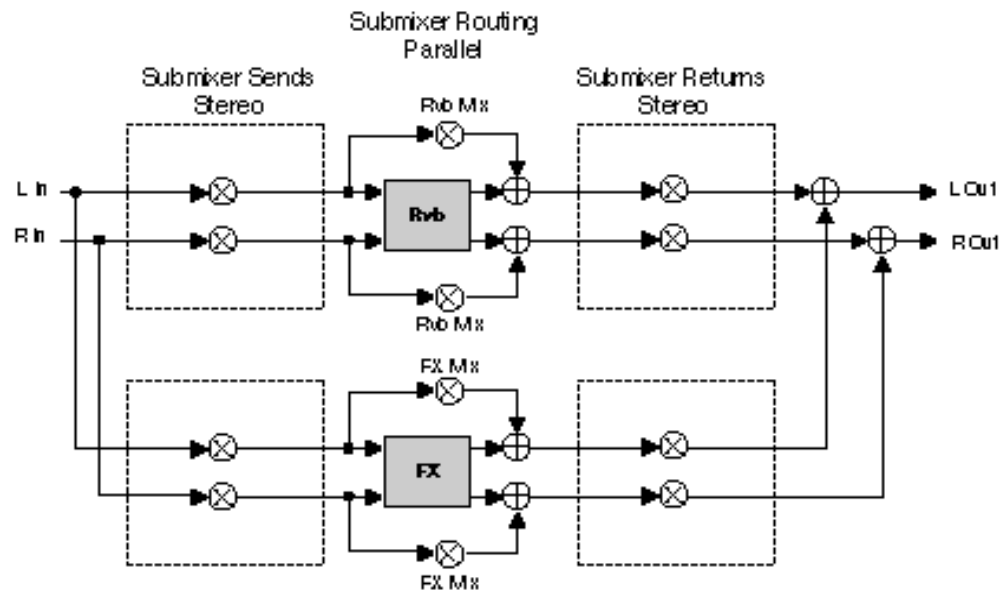
Submixer Sends
1.0 150 Mono

Submixer Returns
1.1 0 Stereo

Submixer Routing
1.2 300 FX into Rvb

Stereo Parallel

- The Rvb and FX blocks process the left and right inputs independently.
- The stereo outputs of both the Rvb and FX blocks are combined at the main outputs.



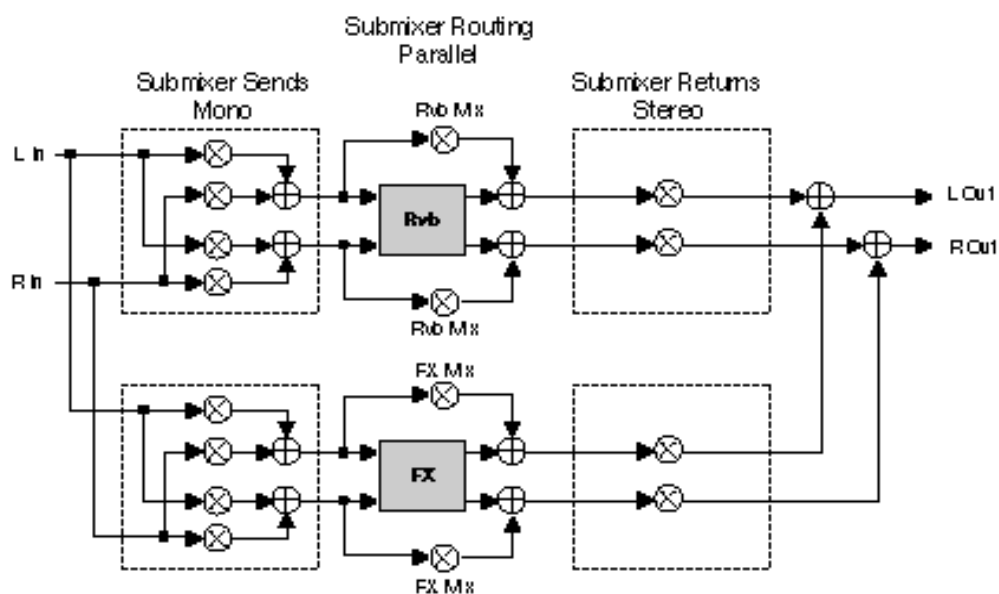
Submixer Sends
1.0 0 Stereo

Submixer Returns
1.1 0 Stereo

Submixer Routing
1.2 0 Parallel

Mono In/Stereo Out Parallel

- The Rvb and FX blocks each independently process a mono mix of the left and right inputs.
- The stereo outputs of both the Rvb and FX blocks are combined at the main outputs.



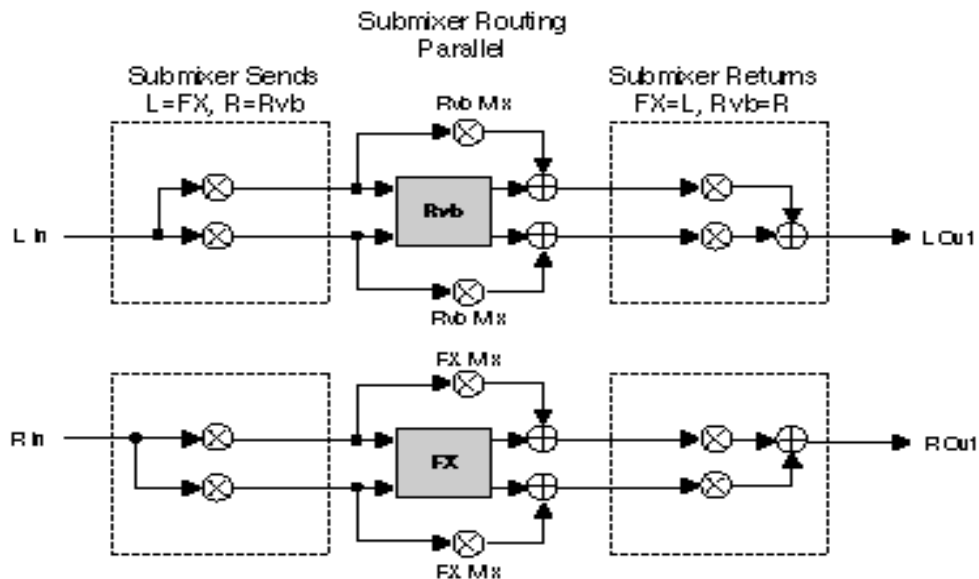
Submixer Sends
1.0 150 Mono

Submixer Returns
1.1 0 Stereo

Submixer Routing
1.2 0 Parallel

Dual Mono In/Stereo Out

- The Rvb block processes only the left input as a mono signal, and the FX block processes only the right input as a mono signal.
- The stereo outputs of both the Rvb and FX blocks are combined at the main outputs.



Submixer Sends

1.0 100 L=Rvb, R=FX

Submixer Returns

1.1 0 Stereo

Submixer Routing

1.2 0 Parallel